

Venture Capital and Entrepreneurial Management

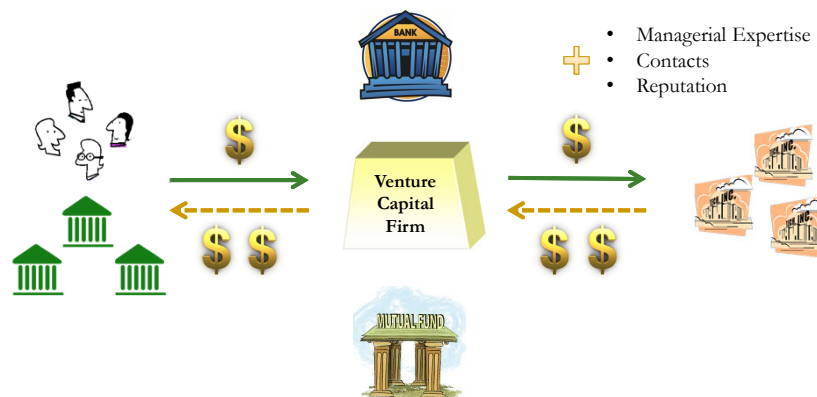
MGMT 264

VC Firm Structure and Activities

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Venture Capital Firms

- Venture Capital (VC) firms are **financial intermediaries** that provide **equity financing** to **privately-held enterprises**.



Founders are willing to give from 10-14% more equity to VC if they are more reputable, like sequoia.

"You pass from being the weird guy doing weird things to the guy backed by goldman"

Venture Capital Firms

- Offers made by VCs with a high reputation are **three times** more likely to be accepted, and high-reputation VCs acquire start-up equity at a **10-14% discount**. Why?

Dependent Variable = VC Offer Accepted N = 148 Observations				Dependent Variable = Relative Valuation Offered N = 148 Observations			
Independent Variables	(5-1)	(5-2)	(5-3)	Independent Variables	(6-1)	(6-2)	(6-3)
High industry deal experience	1.080** (0.512)	1.168** (0.532)	1.068** (0.556)	High industry deal experience	-0.135** (0.057)	-0.144** (0.060)	-0.1098*** (0.036)
High network resources rating			1.415** (0.519)	High network resources rating			-0.048 (0.045)
Relative valuation offered		0.606 (0.887)	2.157 (1.425)	VC offer accepted		0.026 (0.050)	0.041 (0.032)
Angel investor			0.558 (0.799)	Angel investor			-0.021 (0.084)
Corporate VC			-0.450 (0.972)	Corporate VC			0.016 (0.110)
L financing offered			-0.329 (0.593)	L financing offered			0.261*** (0.054)
Equity taken threshold			1.015 (0.851)	Equity taken threshold			-0.291*** (0.067)
Start-up fixed-effects	Yes	Yes	Yes	Start-up fixed-effects	Yes	Yes	Yes
Prob > χ^2	0.028	0.070	0.016	Constant	0.885*** (0.025)	0.889*** (0.027)	0.832*** (0.090)
Log likelihood	-49.195	-48.962	-42.974	Adj. R ²	0.098	0.094	0.460

** or *** indicate statistical significance at the 5% or 1% level, respectively.

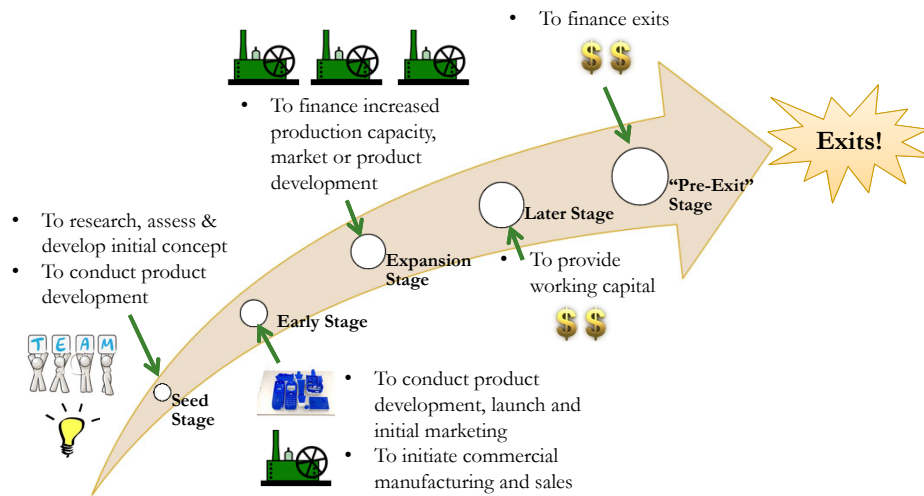
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Source, Hsu, JOF (2004)

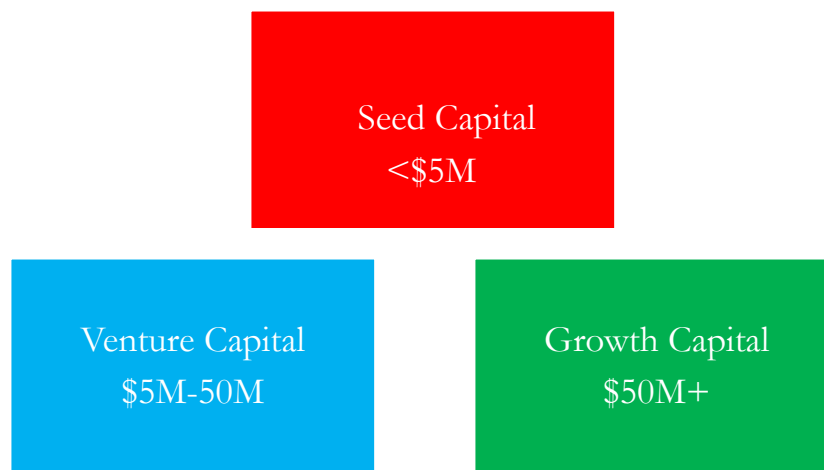
Key Difference Between Early and Late Stages

- Seed
 - Start-up
 - Early Stage
 - Second/third round
- } Not Leveraged
- Expansion/development
 - Bridge/mezzanine
 - MBO, MBI, BIMBO...
- } Often Leveraged

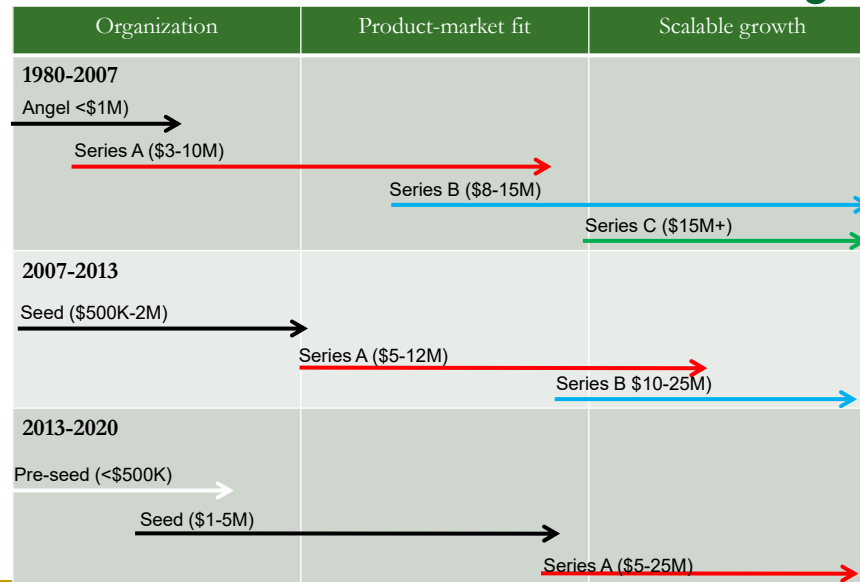
Investment Stages



Segmentation of the Industry



VC Investment Model Shifts to the Right



Venture Capital

- Intriguing Facts

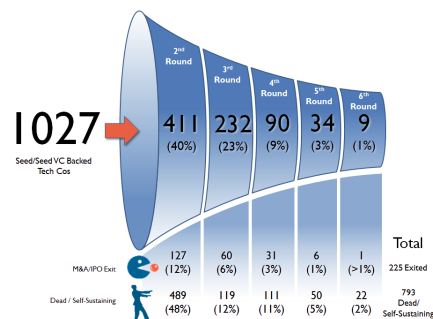
- The success rate is still low given the thin selection funnel...

Figure: The VC Selection Funnel



Source: Goldman Sachs Merchant Banking

Figure: The VC Funding/Exit Funnel

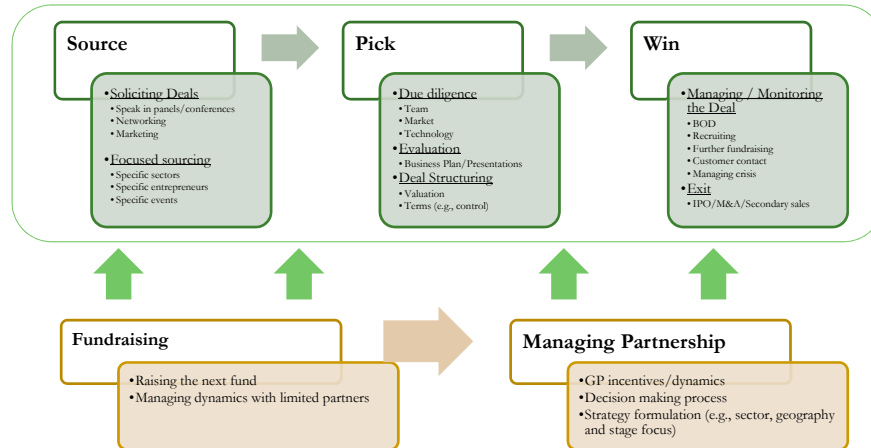


*Based on data from a cohort of tech companies that raised seed funding in 2009 and 2010.

Source: CBInsights.com

Venture Capital Firms

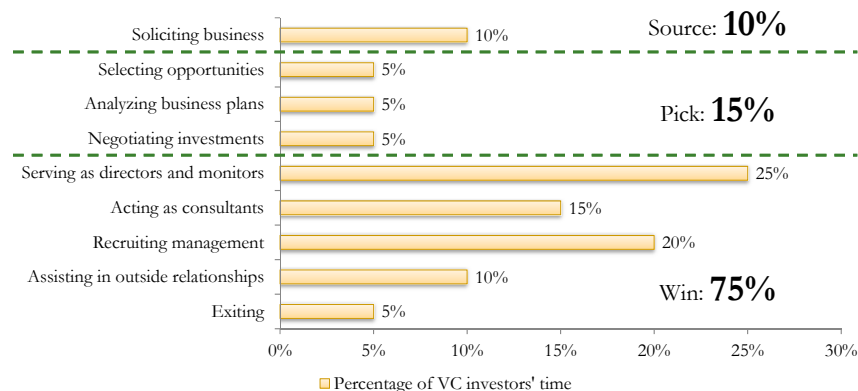
- Activities



Venture Capital Firms

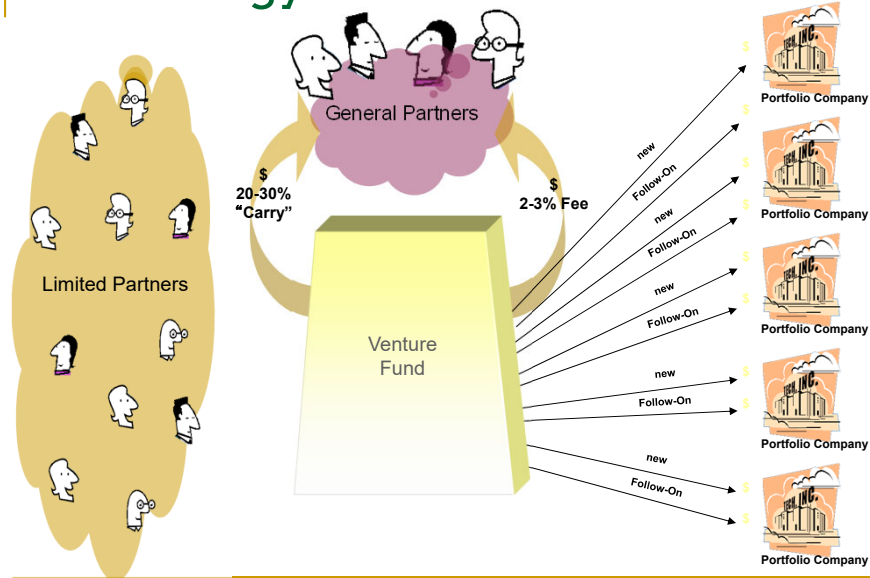
- Activities

■ How do VC investors spend their time?



Source, Zider, Harvard Business Review (1998)

The Ecology of a Venture Fund



Venture Capital Firms

■ Heterogeneity and Persistence of Fund Performance

Sample	Equal Weighted			Size Weighted			
	All Funds	VC Funds	Buyout Funds	All Funds	VC Funds	Buyout Funds	
IRR _{VG5}	0.11	0.10	0.12	0.13	0.15	0.12	Median
IRR after 5 years of fund existence	0.18	0.20	0.14	0.18	0.30	0.13	Standard Deviation
	(0.36)	(0.40)	(0.24)	(0.32)	(0.46)	(0.23)	
	[0.02;0.22]	[0.02;0.21]	[0.02;0.22]	[0.03;0.23]	[0.04;0.42]	[0.02;0.21]	

	Dependent Variable: PME (Public Market Equivalent)							
	Full Sample				VC Only		Buyout Only	
PME _{t-1}	0.54	0.49	0.46	0.69	0.62	0.17	0.19	
	(0.17)	(0.21)	(0.21)	(0.21)	(0.25)	(0.08)	(0.06)	
PME _{t-2}	0.28	0.39	0.28	0.48	0.07	0.07	0.05	
	(0.13)	(0.14)	(0.13)	(0.20)	(0.20)			
PME _{t-3}			0.32					
			(0.24)					
log(Size)			0.09					
			(0.05)					
log(Sequence)			0.20					
			(0.20)					
VC Dummy	0.41	0.51	0.46	0.49	0.60			
	(0.12)	(0.16)	(0.14)	(0.20)	(0.17)			
Firm FE	No	No	No	No	No	No	No	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.19	0.21	0.20	0.25	0.27	0.12	0.23	0.10
No. of Observations	398	225	225	128	225	323	184	41

Key Findings:

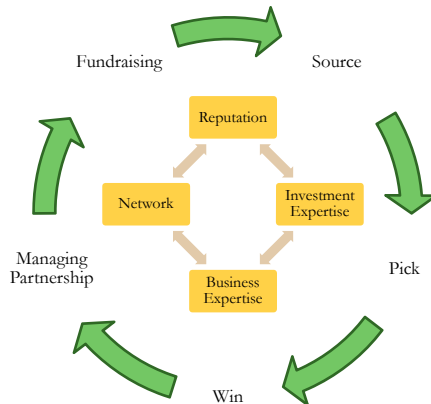
- Substantial heterogeneity across funds exists.
- Return persist strongly across subsequent funds of a partnership.

WHY?

Source: Kaplan & Schoar, JOF2005

Venture Capital Firms

- The Virtuous Circle

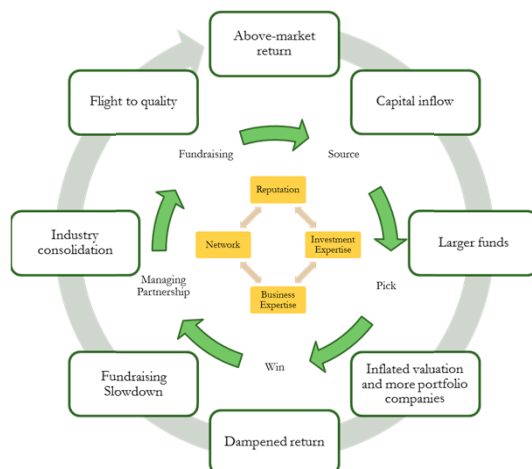


The Capability Inquiries

- Where?
 - At the Partner level
 - At the Organization level
- Scalable?
 - Codified vs. Tacit knowledge
- Transferable?
 - Context-specific vs. Generic resources
 - Boundary of network

Venture Capital Firms

- Strategies and Capabilities



The Strategy Inquiries

- What?
 - Sectors/stage and geography focus?
 - Deals to pick?
 - Capabilities to grow?
- When?
 - Scale fund size and add partners?
 - Expand (e.g., other sectors, stage and geography)

Venture Capital Funds

- Fundraising

■ Limited Partners

- Endowments
- Foundations
- Family Offices



- Pension Funds
- Sovereign Wealth Funds
- Insurance
- FoF



- Track Record
- Partners
- Investment Strategies
- **Geography**



- Deployable capital
- Time horizon
- Long-term commitment
- Value-add
- **Culture fit**



General Partners

Characteristics of Venture Capital

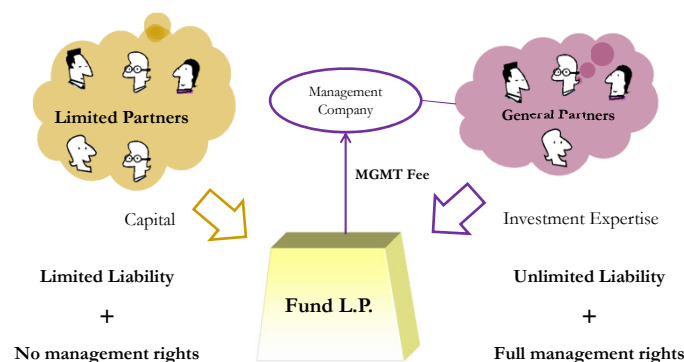
Legal Structure of the Institutional VC Firm

The Venture Capital Firm

- Entity Legal Structure
 - Ownership: Private
 - Legal structures
 - Limited Partnership (very typical for an institutional VC)
 - Limited Liability Corporation
 - Corporation
 - Ownership: Corporate (e.g., Intel Capital, Google Ventures, Comcast Ventures, Qualcomm Ventures)
 - Both financial and strategic objectives
 - Contribution to earnings of parent
 - Insights into relevant industries
 - Changing business model: increased behavior as financial investor
 - Lead deals, set valuation, funding for follow on investments

The Venture Capital Fund - Legal Structure

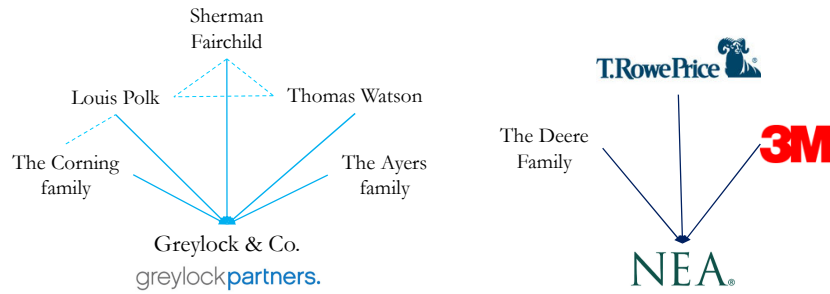
- Limited Partnership (L.P.)



Venture Capital Funds

- Limited Partners

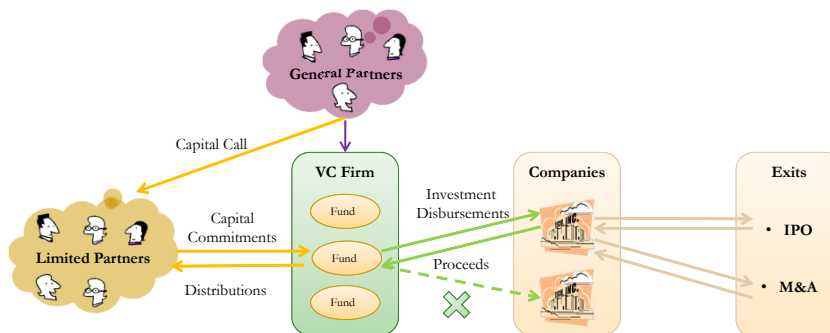
- Selected limited partners of Greylock Partners and New Enterprises Associates (NEA) in the early days



Source: HBS Cases 9-813-002 & 9-813-097

Venture Capital Funds

- Structure



Fundraising:

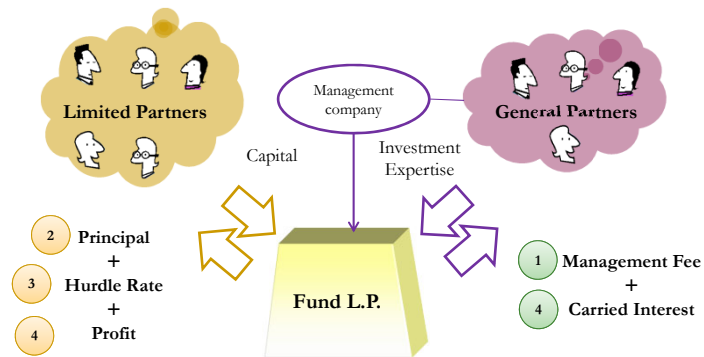
- Committed capital vs. Paid-in capital
- GPs' strategy of calling capital from LPs

Distribution:

- Distribution is often made on a deal-by-deal basis
- Norm: Proceeds are not re-deployable for GPs

Venture Capital Funds

- Economic Distribution



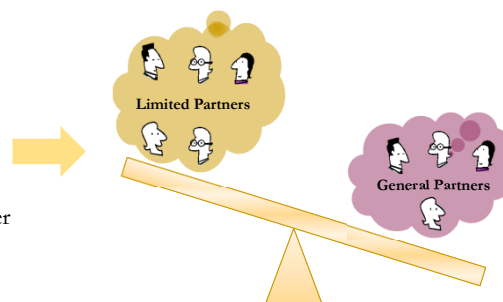
The Venture Capital Firm

- Fundamental Issues in LP/GP dynamics

"Signing up with a private equity firm for 10 years is like being in the jungle with a lion on top of you. You have to hope it's a friendly lion."

---- A limited partner

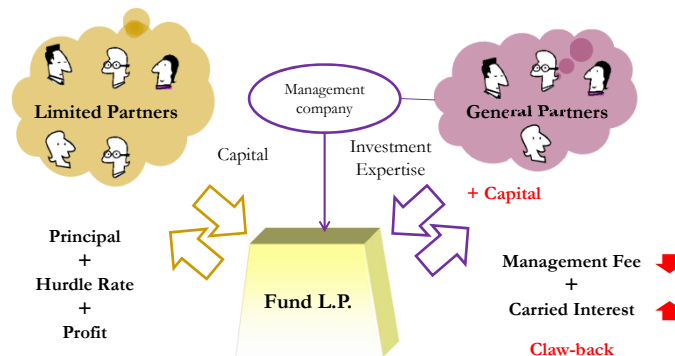
Information Asymmetry between LPs and GPs



The Venture Capital Firm

- Fundamental Issues in LP/GP dynamics

■ Alignment of Interest



Venture Capital Funds

- Economic Distribution

■ A Numerical Example:

□ Assumptions

- Committed & paid in capital by the LP = \$10.0M (assume: capital was paid in on day 1)
- All the capital is deployed in one deal
- Hurdle Rate = 8%
- Carry interest = 20%
- Management Fee = 2% (annual)
- A liquidity event occurs at the end of Year **10**, which returns **\$100.0M** of cash to the Venture Capital fund.

Venture Capital Funds

- Economic Distribution

■ The Waterfall Analysis

	Limited Partners	General Partners
Management Fee		\$2.0M [=10 x 2% x \$10.0M]
Principal	\$10.0M	
Interest (Hurdle)	\$11.6M [= \$10.0M x 1.08 ¹⁰ - \$10M]	
Profit Sharing	\$62.7M [= (\$100.0M - \$11.6M - \$10.0M) x 80%]	\$15.7M [= (\$100.0M - \$11.6M - \$10.0M) x 20%]
Total	\$84.3M	\$17.7M
IRR	24%	

- **Question:** Total distribution = \$84.3M + \$17.7M = \$102.0M > \$100.0M. Why?

Venture Capital Funds

- Economic Distribution

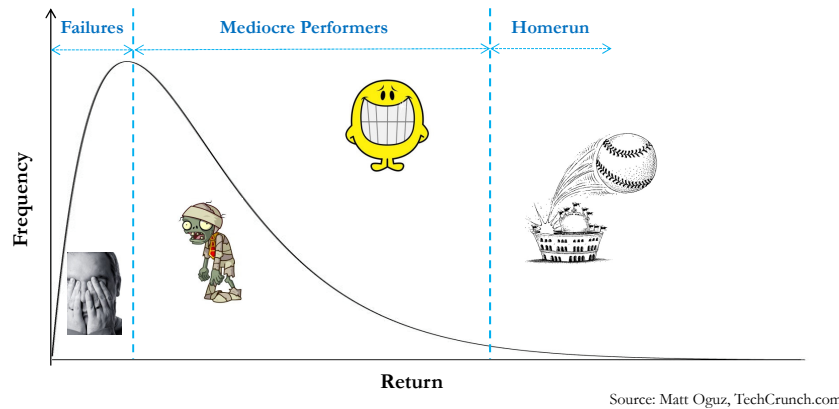
■ Catch-up (Claw-back) Clause for GPs

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Principal	\$10.0M	
Interest (Hurdle)	\$11.6M [= \$10.0M x 1.08 ¹⁰ - \$10M]	
Profit Sharing	\$62.7M [= (\$100.0M - \$11.6M - \$10.0M) x 80%]	\$15.7M [= (\$100.0M - \$11.6M - \$10.0M) x 20%]
Total	\$84.3M	\$17.7M
% of Profit Shared	82.6% >80% [= (\$84.3M - \$10.0M) / (\$100.0M - \$10.0M)]	17.4% <20% [= \$15.7M / (\$100.0M - \$10.0M)]
Catch-up	\$82.0M [= \$84.3M - 2.6% * \$90.0M]	\$20.0M [= \$17.7M + 2.6% * \$90.0M]

Fundamental Issues

- Return and Risk

■ Return Distribution of VC firms' Investments



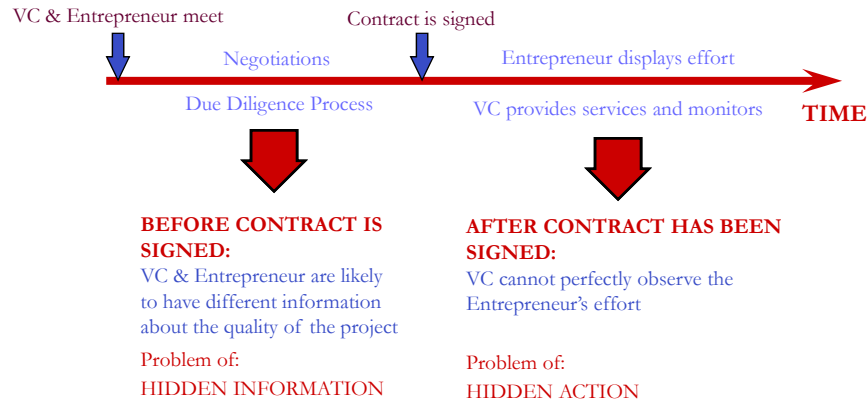
Ten Criteria Most Frequently Rated "Essential" by VCs

■ Capable of sustained intense effort	64%
■ Thoroughly familiar with market	62%
■ At least five times return in five years	50%
■ Demonstrated leadership in the past	50%
■ Evaluates and reacts well to risk	48%
■ Investment can be made liquid	44%
■ Significant market growth	43%
■ Track record relevant to venture	37%
■ Articulates venture well	31%
■ Proprietary protection	29%

Source: Block and MacMillan

Fundamental Issues

- Information and Agency Problems



Fundamental Issues

- The Problem of Hidden Information

■ CLASSIC EXAMPLE

The 'lemon' problem



■ ASSUME

Buyers of used cars can only assess average quality of cars on the market; sellers know exact quality

■ WHAT HAPPENS?

Si el mercado paga la media por un coche, si tienes un coche bueno no lo vendes, porque t pagan menos de lo q vale. Devtal manera que decrece progrsvivamente la calidad el mercado

*Strategy and patents
You file 15-20 patents while only one matters, because patent info is public. That way people do not know which one is you key one

Fundamental Issues

- Implications of Hidden Information for Venture Capital

■ ASSUME

Entrepreneur knows project quality; venture capitalist does not.

■ WHAT HAPPENS?

■ POSSIBLE COUNTER MEASURES?

Fundamental Issues

- The Problem of Hidden Action

■ CLASSIC EXAMPLE

Car Insurance



■ ASSUME

The actions of car owners after a contract is signed cannot be perfectly observed by insurance companies.

■ WHAT HAPPENS?

Fundamental Issues

- Implications of Hidden Action for Venture Capital

■ ASSUME

Venture capitalist cannot perfectly observe the entrepreneur's effort.

■ WHAT HAPPENS?

■ POSSIBLE COUNTER MEASURES?

Takeaway

■ Venture Capital Firm

- Working Definition
- Stage focus

■ Venture Capital Activities

- Source
- Pick
- Win

■ Fundamental Issues

- Return Distribution
- Time Horizon
- Information Asymmetry between VCs and Entrepreneurs

Next Class:

Opportunity Evaluation

Your Questions?

